# Verification and Progress on Purity of Pure

## A machine-checked report on Goodman's project notes

27 July 2026; updated 29 July 2026

**Purpose.** This report records what we have verified in Jeremy Goodman's July 2026 project notes on the consistency of Purity of Pure with Bacon's logical combinatorialism. It has two aims. First, it separates the claims that have been verified exactly from claims that require qualifications, repairs, or further formulation. Second, it states the new theorems proved during the formalization that genuinely narrow the open consistency problem.

**Bottom line.** Goodman's central consistency question remains open. Isabelle nevertheless settles the determinate subsidiary claims in the notes: each is proved, refuted and repaired, or assigned the explicit qualification under which it is true. In particular, the formalization identifies and repairs a modal gap in T3 and a false countability argument in M5; verifies the four T6 contradiction routes and the T9 cardinal dichotomy from their explicit premises; completes the rebuilt M5 model containing the displayed exotic operator; and completes the denotational step in Bacon's footnote-59 argument. In Bacon's exact tree semantics, the displayed footnote-59 term denotes the required diagonal, with identity interpreted at the appropriate higher types. Consequently no interpretation in that semantic framework can validate $\mathsf{PP}$ together with the $\mathsf{PP}$-free M1 background and $\mathsf{QSS}$. This rules out completion of the present tree-model program, although it does not prove that the abstract theory has no other Henkin model.

The formalization also proves that Recombination together with zeroary Exhaustion and unique proposition-level fundamentality supplies the $\mathsf{fun}'$ witness used in T6. Hence T6 has the following exact semantic consequence: any model of the repaired background theory plus $\mathsf{PP}$ must falsify global $\mathsf{L2}$ or global $\mathsf{TU}$. That consequence sharply constrains a positive answer, but it does not decide whether such a model exists. We have also settled Goodman's proposed calibration of $\mathsf{L2}$ in Bacon's appendix model: a closed logical operator $Z$ refutes $\mathsf{L2}$ when the pure unary operators include the denotations of closed terms containing only logical vocabulary, are closed under composition, and admit a $\mathsf{fun}'$ proposition. Thus the counterexample persists through every enlargement satisfying those conditions, including any prospective $\mathsf{PP}$ enlargement on Bacon's tree frame that validates the repaired background theory. This is a substantive result about the notes, although it does not construct the missing interpretation of $\mathsf{Pure}$ and therefore does not answer the consistency question.

# 1 Formal setting and verification standard

The formal development represents Bacon's typed language and proof system directly in Isabelle/HOL. It includes Bacon's system $H$, classical and Boolean identity principles, the Rule of Equivalence, vector Equivalence, the definition of necessity by identity with truth, and the additional purity/fundamentality vocabulary used in Goodman's notes. The development distinguishes object-language derivability from validity in a particular model. This matters throughout: a semantic argument valid in Bacon's appendix model is not thereby a derivation in the theory, and an Isabelle theorem with explicit premises does not show that those premises hold in the intended model.

The complete audit checks the theorem objects supporting the claims summarized below. It presently checks 136 theorem objects, and fresh builds pass. None of these proofs contains an admitted lemma, an untrusted proof step, or an undischarged assumption beyond the assumptions stated in the theorem itself. The type restriction carried by the general cardinal theorems in T9 is part of their formally checked statement, not a further mathematical premise.

The semantic evaluation used below interprets identity at proposition type, the type of unary operators, and the next classifier type in their respective domains. In particular, identity between unary operators is pointwise identity over the proposition domain; it is not obtained by first treating higher-type values as propositions. This correction to an auxiliary evaluator preserves all earlier verified results and is essential to the exact M1 calculation.

# 2 Background definitions and two important distinctions

At proposition type, Goodman's definable surrogate for fundamentality is

$$\mathsf{fun}'(p) \quad \stackrel{\text{def}}{=} \quad \forall X\, Y\big((\mathsf{Pure}(X)\,\&\,\mathsf{Pure}(Y)\,\&\,Xp = Yp) \to X = Y\big).$$

Let $G$ be the pure reversible unary operators, and let

$$X \approx Y \quad \stackrel{\text{def}}{=} \quad \exists Z \in G\ (X = Y \circ Z).$$

Weak $\mathsf{L2}$ says that if $X$ and $Y$ are pure, $p$ and $q$ are $\mathsf{fun}'$, and $Xp = Yq$, then $X \approx Y$.

Two distinctions uncovered by the formalization are central.

**The scope of QLN.** Goodman's general formulation requires the fundamental arguments of a QLN instance to be pairwise distinct. The present problem stipulates exactly one fundamental entity, which is a proposition. Hence every instance of arity at least two is vacuous. The zeroary and unary instances are therefore the complete substantive QLN package for this problem; no higher-arity instance has been silently omitted.

**Recombination versus Exhaustion.** Unary Recombination proves a pointwise modal identity principle. It does not, by itself, permit necessitating the universal closure required by the usual QSS derivation. Zeroary Exhaustion supplies exactly that missing step. With unique proposition-level fundamentality, the repaired package derives the existence of a $\mathsf{fun}'$ proposition.

**The scope of Modalized Functionality.** Bacon–Dorr's Intensionality argument, as formalized in CEV, proves Modalized Functionality for operators whose result type is proposition:

$$X, Y : \sigma \to t.$$

The argument type $\sigma$ is arbitrary. The broader two-type formula for $X, Y : \sigma \to \tau$, with arbitrary result type $\tau$, remains a separate schema and a direct model obligation. The T2–T8 arguments formalized here use the proposition-valued instances and are unaffected by this distinction. What is not justified is the earlier suggestion that the entire arbitrary-result schema is already redundant over bare CEV.

**Conditional model-theoretic results.** A formally verified theorem may still have explicit premises concerning Bacon's function domains, invariance, or QSS. We therefore distinguish a verified conditional theorem from a proof that all of its premises hold in Bacon's particular appendix model.

## 3 Goodman's L2 principles

Goodman's July notes state one principle bearing the label $L$, in two forms. Weak L2 is

(L2) $$\mathsf{Pure}(X) \,\&\, \mathsf{Pure}(Y) \,\&\, \mathsf{fun}'(p) \,\&\, \mathsf{fun}'(q) \,\&\, Xp = Yq \quad \Longrightarrow \quad X \approx Y.$$

The strong form requires the same reversible operator to witness both the relation between the operators and the relation between their inputs:

(strong L2)

$$\mathsf{Pure}(X) \,\&\, \mathsf{Pure}(Y) \,\&\, \mathsf{fun}'(p) \,\&\, \mathsf{fun}'(q) \,\&\, Xp = Yq \quad \Longrightarrow \quad \exists Z \in G\,(X = Y \circ Z \,\&\, q = Zp).$$

The notes contain no separate principle labeled $L1$. Accordingly, the plural “L principles” below refers to weak and strong $\mathsf{L2}$.

### 3.1 Object-language verification

Both forms have been encoded as closed, well-typed object-language formulas. Isabelle verifies all four uses made in T6:

$$T_0 + \mathsf{PP} + \exists\mathsf{fun}' + \mathsf{L2} + \mathsf{Inv} \vdash \bot,$$
$$T_0 + \mathsf{PP} + \exists\mathsf{fun}' + \mathsf{L2} + \mathsf{WI} \vdash \bot,$$
$$T_0 + \mathsf{PP} + \exists\mathsf{fun}' + \mathsf{L2} + \mathsf{TU} \vdash \bot,$$
$$T_0 + \mathsf{PP} + \text{strong } \mathsf{L2} + \mathsf{RS} \vdash \bot.$$

The repaired Recombination–Exhaustion bridge derives the required $\exists\mathsf{fun}'$ premise from Recombination, zeroary Exhaustion, and unique proposition-level fundamentality. Isabelle also verifies the uses of $\mathsf{L2}$ in T7–T9, including uniqueness of kind in T8 and the conditional cardinal dichotomy in T9.

### 3.2 Refutation of L2 in Bacon’s appendix model and its enlargements

Goodman’s first suggested attack asks whether $\mathsf{L2}$ holds in Bacon’s appendix model when the pure unary operators are exactly the denotations of closed terms containing only logical vocabulary. Isabelle answers this question negatively, and proves a stronger result for enlargements of those operators.

Define the unary operator

$$Z(P) = \{\, w : P(w^\frown 0) \text{ and } P(w^\frown 1) \text{ have different truth values} \,\}.$$

There is a closed term containing only logical vocabulary whose denotation is exactly $Z$. Hence $Z$ is one of the pure unary operators denoted by closed terms containing only logical vocabulary. Isabelle proves:

(i) $Z$ is surjective;

(ii) $Z(P) = Z(\neg P)$, so $Z$ is not injective;

(iii) for every composition-closed class of pure unary operators containing $Z$, surjectivity of $Z$ carries each $\mathsf{fun}'$ proposition $p$ to the $\mathsf{fun}'$ proposition $Zp$;

(iv) $Z \notin G$.

Surjectivity is witnessed explicitly by

$$P_S = \{ v^\frown 0 : v \in S \}, \qquad Z(P_S) = S.$$

For the closed-logical operators themselves, clause (iii) yields the earlier right-cancellation result. More generally, let $S$ be a class of pure unary operators containing the closed-logical operators and closed under composition. Assume that $S$ admits a $\mathsf{fun}'$ proposition $p$, where $\mathsf{fun}'$ is understood relative to $S$. Since $Z$ is surjective, clause (iii) shows that $Zp$ is also $\mathsf{fun}'$ relative to $S$, and

$$\mathsf{id}(Zp) = Zp = Z(p).$$

If L2 held relative to $S$, identity and $Z$ would therefore be of the same kind. A reversible witness for that claim would supply a left inverse for $Z$, making $Z$ injective. Clause (ii) rules this out. Consequently weak L2 fails relative to $S$. Since strong L2 entails weak L2, it fails as well.

## 3.3 What this advances

This settles the first open problem in Goodman's list: L2 is false in Bacon's appendix model under the proposed interpretation of purity. It also shows that enlarging the pure unary operators cannot restore L2 on the same tree frame, provided that the enlarged class remains closed under composition and admits a $\mathsf{fun}'$ proposition. The earlier Isabelle analysis had shown that no counterexample occurs among identity, necessity, possibility, constant truth, and constant falsity; $Z$ is the first verified nonreversible closed logical operator that nevertheless preserves the distinctions relevant to $\mathsf{fun}'$.

The result does not prove the consistency of PP. Bacon's appendix model still fails PP at the unary-operator type, and the enlargement theorem assumes rather than constructs a suitable enlarged interpretation of the pure unary operators. What it shows is that failure of L2—one of the two escape routes forced by T6—persists throughout the relevant enlargements of Bacon's semantic setting. It also shows that a negative solution cannot treat L2 as automatic merely because purity includes all closed logical denotations or because PP is added on the same tree frame.

## 3.4 Automated proof search for L2 from PP

We also used Vampire to test whether a finite representation of $T_0 + \mathsf{PP} + \exists\mathsf{fun}'$ yields weak L2. The first experiment presented the entire represented stock together with the denial of L2 as one higher-order refutation problem. It returned no proof after one hour.

We then divided the proposed derivation into separate intermediate claims. Suppose that $X$ and $Y$ are pure, $p$ and $q$ are $\mathsf{fun}'$, and $Xp = Yq$. Vampire proves that $X$ and $Y$ then agree on all equations obtained by composition on the left by pure operators:

$$(*) \qquad \forall A\, B\, (\mathsf{Pure}(A)\,\&\,\mathsf{Pure}(B) \implies (A \circ X = B \circ X \leftrightarrow A \circ Y = B \circ Y)).$$

It also proves each of the following conditional routes to $\mathsf{L2}$:

1. if pure $X$ and $Y$ satisfying $(*)$ must satisfy $X \approx Y$, then $\mathsf{L2}$;
2. if every two $\mathsf{fun}'$ propositions $p, q$ satisfy $q = Zp$ for some $Z \in G$, then $\mathsf{L2}$;
3. even the weaker condition that, for every pure $Y$ and every two $\mathsf{fun}'$ propositions $p, q$, there is some $Z \in G$ such that $Yq = Y(Zp)$, implies $\mathsf{L2}$.

The remaining question is whether $T_0 + \mathsf{PP} + \exists\mathsf{fun}'$ proves any one of these three sufficient conditions. We tested each condition separately, first with genuine higher-order function types and then in a multisorted first-order representation in which unary operators are reified as objects. With a 60-second, six-core search for each conjecture, all six runs timed out. None returned either a proof or a countermodel.

These experiments establish no nonderivability result. Their positive value is diagnostic: Vampire rapidly proves the algebraic steps from each proposed intermediate condition to $\mathsf{L2}$, but it does not presently obtain any of those conditions from $\mathsf{PP}$. Thus the automated search isolates the same mathematical gap as the formal analysis: a negative argument along this route requires a substantive new consequence of $\mathsf{PP}$, rather than merely a longer search through the already identified composition and cancellation steps.

# 4 Audit of the object-language theorems

| Item | Status | Result |
|---|---|---|
| T1 | Verified | Under zeroary Exhaustion, every pure proposition is truth or falsity. Consequently the biconditional operators are identity and negation, and $\mathsf{WI}$ collapses to $\mathsf{Inv}$. |
| T2a | Verified | $\mathsf{fun}'$ is closed under every pure reversible operator, hence under negation. |

| Item | Status | Result |
|---|---|---|
| T2b | Verified | Truth and falsity are not $\mathsf{fun}'$. A $\mathsf{fun}'$ witness is distinct from truth, falsity, and its negation. |
| T2c | Verified, stronger | Every pure proposition is possibly identical to a $\mathsf{fun}'$ witness. The proof does not use $\mathsf{PP}$. |
| T2d | Verified, stronger | Possible purity of the witness follows without Persistence, although Goodman's displayed route uses it. |
| T2e | Verified | The proposition that the witness is noncontingent is false but possible. |
| T2f | Verified conditionally | The six displayed propositions are pairwise distinct under the explicit $\mathsf{fun}'(r)$ antecedent. The antecedent is essential. |
| T3 | Repaired | The advertised proof has a modal gap: it reaches possible identity where identity is required. The theorem follows after adding zeroary Exhaustion, or from a weaker pure-identity rigidity principle. A two-world modal abstraction verifies that the advertised intermediate principles do not by themselves close the gap. |
| T4 | Verified, stronger | For each closed pure $C : t \to (t \to t)$, $\sim\mathsf{fun}'_{t\to t}(C(r))$. The proof uses only the purity schema and application closure; neither $\mathsf{PP}$ nor $\mathsf{fun}'(r)$ is needed. |
| T5 | Verified conditionally | The assumptions displayed in the notes, together with $\mathsf{fun}'(r)$, yield a $\mathsf{fun}'$ proposition distinct from $r$ and $\sim r$. This does not establish consistency of the antecedent. |
| T6 | Verified | All four contradiction routes are formalized: $\mathsf{L2{+}Inv}$, $\mathsf{L2{+}WI}$, $\mathsf{L2{+}TU}$, and strong-$\mathsf{L2{+}RS}$. Recombination, the proposition-type instance of Exhaustion, and unique fundamentality derive, rather than assume, the required $\mathsf{fun}'$ witness where appropriate. The advertised T6-$\mathsf{WI}$ master claim is also derived directly from its stated stock, independently of explosion. |
| T7a | Verified | For the liar family, the identity-indexed member is false, while some member indexed by a pure reversible operator is true. |

| Item | Status | Result |
|---|---|---|
| T7b | Underspecified | The notes do not give the kind-level diagonal term, its binders, or precise definitions of “fixed” and “shifted” point. There is no unique theorem to encode. |
| T8 | Verified | The five base kinds are pairwise distinct; L2 gives kind uniqueness; and the assumptions displayed in the notes prove 31 pairwise-distinct pure unary operators and 31 pairwise-distinct propositions. |
| T9 | Verified at the counting level | The PC specification makes the map from sets of kinds to pure operators injective. Representation of the members of each kind by members of $G$ gives an injective code of that kind into $G$. These results imply $$\lvert\mathcal{P}(K)\rvert \leq \lvert K \times G\rvert.$$ Classical cardinal arithmetic then gives $$K \text{ finite} \quad \vee \quad \lvert\mathcal{P}(K)\rvert \leq \lvert G\rvert.$$ Instantiating the abstract types uniformly across Goodman’s full typed object language remains separate. |

# 5 Audit of the model-theoretic claims

| Item | Status | Result |
|---|---|---|
| M1 | Exact denotation and tree-semantic exclusion | At proposition type, the pure propositions in Bacon's model are exactly the two invariant propositions, and the operator testing for membership in that class is the noncontingency operator. The exact footnote-59 liar is typed, beta-reduced, and proved pure from PP, Purity of Fun, and application closure. Isabelle now also computes its denotation in Bacon's exact tree semantics. It is precisely Goodman's displayed diagonal, with proposition and operator identity interpreted by the appropriate local equivalence relations. The object-language formula QSS is evaluated in the same semantics, and the resulting diagonal argument is proved without an assumed semantic diagonal premise.<br>It follows that, whenever Bacon's tree semantics validates the PP-free M1 background and QSS, it cannot validate PP. Thus the present tree-model program cannot be completed into the desired model. At the next type, the infinitary join still exists in the full function domain and has exactly the pure unary operators as its extension; PP holds there exactly when that classifier is itself in the next pure stock. Completeness of the function domain supplies the classifier, but the exact footnote-59 theorem shows that it cannot be pure while the other M1 assumptions and QSS are retained. This is a result about Bacon's tree semantics, not yet a model-independent inconsistency theorem for every CEV+ Henkin model. |
| M2 | Verified | Goodman's construction $T \mapsto G_T$ is a bijection between arbitrary sets of propositions and invariant unary operators. Cantor's theorem yields more invariant operators than propositions, so evaluation at one fundamental proposition cannot be injective on all invariants. The invariance reading of purity therefore violates QSS. |

| Item | Status | Result |
|---|---|---|
| M3 | Verified relative to the pure operators | Relative to the chosen stock of pure operators, $\mathsf{fun}'$ is equivalent to freeness against nonzero pure laws. Such propositions have both extreme views. A countable Boolean stock has a free generator, while the resulting $\mathsf{fun}'$ class is topologically meager in the product topology. |
| M4 | Conditional model-theoretic theorem | Preimage heredity is proved. The lifted-branch witness is $\mathsf{fun}'$ and outside the reversible orbit under explicit assumptions: necessitated $\mathsf{QSS}$, function-space membership and invariance of every operator in the pure stock, and presence of identity, zero, and one. The theorem has not yet been instantiated for Bacon's stock of operators denoted by closed terms containing only logical vocabulary. The multiple-fundamental "wide Fun" argument remains prose-level. |

| Item | Status | Result |
|---|---|---|
| M5 | Verified and repaired; unrestricted generalization refuted | The operator $f = G_T$ obtained by exchanging $s$ and $s'$ is an invariant involution violating TU, WI, and Inv. The notes' subsequent countability argument is false: both the orbit and the displayed world-indexed family are countable. We replace it with a uniform diagonal construction. For every candidate $R$, it produces a two-element pair outside every view of $R$, whose proper views are all empty or universal. The induced invariant nonidentity transposition fixes $R$. If it and identity are both included among the pure operators, $R$ ceases to be $\mathsf{fun}'$, proving the corrected pre-rebuild QSS obstruction. The rebuilding question is now settled. Isabelle constructs the least application-closed pure stock containing every closed logical denotation and the displayed $f$, proves that $f$ is a pure self-inverse operator in the modalized function domain, and rebuilds a fundamental proposition that supplies Recombination and $\mathsf{fun}'$-separation at every world. The rebuilt interpretation validates Bacon's Recombination background. It does not validate PP; that would require the classifier of the enlarged pure stock itself to belong to the next pure stock. The displayed collision calculation is also verified: the closed operator $$\lambda p\ (p \leftrightarrow \mathsf{NC}(p))$$ takes the same value at $\mathsf{NC}(r)$ and truth, although $\mathsf{NC}(r) \neq \mathsf{truth}$, whenever $\mathsf{fun}'(r)$. Extending this collision argument uniformly to merely existentially given invertibles is impossible. Isabelle proves in CEV+ that every operator with an existentially supplied two-sided inverse is injective; in particular, the displayed collision operator is not reversible. Conversely, the rebuilt model's exotic operator is a reversible, non-truth-uniform bijection and has no collision. A PP-specific collision argument would therefore require a further condition constraining the action of the inverse witness. |

| Item | Status | Result |
|---|---|---|
| M6 | Conditional model-theoretic theorem | Under the same explicit assumptions—necessitated QSS, membership of every pure operator in Bacon's function domain, and invariance—distinct substitutions are separated by a $\mathsf{fun}'$ proposition. With identity, zero, and one, a strict-inclusion $\mathsf{fun}'$ pair blocks the indicated joint assignment. Single-coordinate arbitrary realization fails independently by countability. These results have not all been instantiated for Bacon's stock of operators denoted by closed terms containing only logical vocabulary. |
| M7 | Verified, with an open instance | The Tarski diagonal lies outside the range of the enumerated closed terms. For arbitrary invariant operators of Goodman's form $G_T$, every proposition is reachable from $r$ if and only if the orbit map $$i \longmapsto i \cdot r$$ is injective. Whether Bacon's glued $r$ has this property is construction-sensitive and open. |
| 10.1 | Verified, qualified | Bacon's rebuilt-family theorem is proved in a set-theoretic formalization of his appendix model. At type Ind, equality is literal identity, and the theorem's hypotheses therefore force the family to be constant; the theorem cannot supply a family of distinct individual denotations. |

# 6 Corrections and clarifications to the notes

## 6.1 The modal gap in T3

The advertised argument uses necessitated QSS and Persistence to move from a possible fundamental role to an injectivity claim. The formal derivation reaches a statement of the form

$$\Diamond(Y = Z),$$

where the desired conclusion requires $Y = Z$. No rule of $H$ licenses that transition. The gap can be repaired in either of two principled ways.

1. Zeroary Exhaustion collapses pure propositions to truth and falsity, which supplies the required rigidity.
2. One may assume only the weaker principle that possible identity between pure operators implies identity.

The unrestricted version of the second principle is false. Thus the repair must remain restricted to the pure case.

## 6.2 The countability error in M5

The notes propose choosing a pair $\{s, s'\}$ outside the orbit of $r$ because the orbit is countable and, it is said, there are uncountably many world-indexed pairs. But Bacon's worlds are finite sequences of natural numbers, hence countable. The displayed family of pairs is therefore countable as well. Countability alone proves nothing.

The replacement is direct diagonalization. Enumerate the views of $R$ by the worlds. Define a proposition $s_R$ whose truth at the singleton $[n+2]$ disagrees with the corresponding singleton's membership in the $n$-th view of $R$, and let

$$s'_R = s_R \cup \{\langle\rangle\}.$$

Membership above a nonactual world depends only on the final symbol of its sequence. Hence every nonactual view of $s_R$ and $s'_R$ is empty or universal, while both members differ from every view of $R$. Exchanging their membership in the set $T_0$ of true propositions yields an invariant nonidentity operator $f_R = G_T$ with

$$f_R(R) = R.$$

If $f_R$ and identity are both included among the pure operators, evaluation at $R$ is not injective. This is exactly the pre-rebuild QSS obstruction the M5 strategy needs.

The separate rebuilding step is also now verified. Instead of retaining this old $R$, the rebuilt model chooses a new fundamental proposition generic for the enlarged application-closed stock containing Goodman's displayed $f$. That proposition again satisfies Recombination and $\mathsf{fun}'$-separation.

### 6.3 The scope of several semantic claims

The M4 and M6 theorems are genuine mathematical results, but at present they are conditional. They assume that every operator counted as pure belongs to Bacon's function domain and is invariant, together with necessitated QSS and, for some results, the presence of identity and the two constant operators. The proofs do not yet instantiate these assumptions for Bacon's complete stock of operators denoted by closed terms containing only logical vocabulary. We therefore do not count them as unconditional verifications of the corresponding claims about Bacon's particular glued model.

## 7 New theorems advancing the consistency question

The audit was not merely defensive. The formalization has produced several theorems not stated in the original notes. These results materially clarify the logical relations among Goodman's central claims.

### 7.1 The exact footnote-59 obstruction

**Theorem 1** (M1 in Bacon's exact tree semantics)**.** *Interpret* Fun *by the unique distinguished fundamental proposition at each world of Bacon's tree semantics. If an admissible interpretation of* Pure *validates the constant-free purity schema, application closure, Purity of Fun, and* QSS*, then it does not validate* PP*.*

The proof has three machine-checked parts. First, Bacon's closed footnote-59 term is evaluated and shown to denote the operator $D$ such that

$$Dp \quad \leftrightarrow \quad \forall X\, q\, (\mathsf{Pure}(X) \,\&\, \mathsf{Fun}(q) \,\&\, p = Xq \;\rightarrow\; {\sim}Xp).$$

Second, object-language QSS is evaluated using the same type-correct identity relation as the rest of the semantics. Third, QSS identifies each relevant pure operator $X$ with $D$, yielding the diagonal contradiction. Thus the theorem does not assume a separate semantic diagonal premise.

This materially changes the status of the model construction. The verified fragments remain correct, but no enlargement within Bacon's exact tree semantics can reach the complete target while retaining the stated M1 background and QSS. A positive answer therefore requires a different Henkin interpretation. A negative answer would follow if this argument could be generalized to every admissible CEV+ Henkin model.

### 7.2 Recombination, Exhaustion, and the existence of a $\mathsf{fun}'$ proposition

**Theorem 2.** *Unary Recombination yields the pointwise* $\mathsf{QSS}$ *core. Adding zeroary Exhaustion yields the necessitated universal form needed for* $\mathsf{QSS}$. *Together with unique proposition-level fundamentality, the resulting stock proves that a* $\mathsf{fun}'$ *proposition exists.*

This theorem removes the previously assumed existential $\mathsf{fun}'$ premise from the premises used in T6. It also makes explicit exactly where Exhaustion, rather than Recombination, enters the object-language argument.

### 7.3 An exact semantic consequence of T6

**Theorem 3.** *Any model validating Bacon's proof theory, Recombination, the proposition-type instance of Exhaustion, unique proposition-level fundamentality, and* $\mathsf{PP}$ *must falsify global* $\mathsf{L2}$ *or global* $\mathsf{TU}$*:*

$$\sim \mathsf{GValid}(\mathsf{L2}) \quad \vee \quad \sim \mathsf{GValid}(\mathsf{TU}).$$

*Moreover, some world in the constructed model falsifies one of the two closed formulas.*

Here $\mathsf{GValid}(A)$ means that the closed formula $A$ is true at every world of the model. This formalizes the central dilemma in Goodman's notes. It does not select a disjunct. A positive model may exploit either cross-kind collisions on $\mathsf{fun}'$ inputs or a pure reversible operator that is neither uniformly truth-preserving nor uniformly truth-flipping.

### 7.4 The rebuilt M5 model

**Theorem 4** (Completion of the rebuilding step)**.** *There is a typed interpretation of Bacon's Recombination background whose pure denotations form the least application-closed stock containing all closed logical denotations and Goodman's displayed exotic operator $f$. That stock is countable. The operator $f$ is in the modalized unary function domain, commutes with taking views, is self-inverse, is not truth-uniform, and is not a biconditional operator. A rebuilt fundamental proposition supplies Recombination and* $\mathsf{fun}'$*-separation at every world.*

This discharges the rebuilding step that Goodman's notes left "modulo Theorem 10.1." It verifies the intended M5 independence result for Bacon's Recombination background. It does not provide a model of $\mathsf{PP}$: $\mathsf{PP}$ would add the further requirement that the classifier of this enlarged pure stock itself occur at the next type.

### 7.5 A uniform pre-rebuild M5 obstruction

**Theorem 5** (Uniform version of the M5 construction)**.** *For every candidate proposition $R$, there is a pair $\{s_R, s'_R\}$ disjoint from the set of all views of $R$. Every nonactual view of either member is empty or universal. The associated invariant transposition fixes $R$ and is not identity.*

This is stronger and cleaner than the original M5 selection argument. It shows that an exotic invariant operator fixing any preassigned $R$ can be constructed without cardinal assumptions. It shows that the pre-rebuild failure is caused by keeping the old fundamental proposition fixed while enlarging the pure operators. The preceding theorem shows how rebuilding the fundamental proposition avoids that failure.

### 7.6 The T9 and M7 reductions

**Theorem 6** (T9 dichotomy)**.** *If* PC *supplies an injection $\mathcal{P}(K) \to P$ into the pure unary operators, and every $\approx$-class injects into $G$, then*

$$K \text{ is finite} \quad \vee \quad |G| \geq 2^{|K|}.$$

**Theorem 7** (Invariant reachability)**.** *Every proposition is reachable from $r$ by an invariant operator if and only if*

$$i \longmapsto i \cdot r$$

*is injective.*

The second theorem turns the open invariant form of Fundamental Completeness into a precise property of the chosen fundamental proposition. It also confirms that the issue depends on Bacon's particular construction rather than following from general facts about substitutions.

## 8 Where the open question now stands

Goodman's question asks whether Bacon's background theory, unique proposition-level fundamentality, and PP have a model. The verified results clarify both possible answers without deciding between them.

**Affirmative route.** A positive answer requires a model of the full theory plus PP, with purity interpreted at every relevant type. T6 shows that such a model must falsify L2 or TU.

Thus it must contain either pure operators from distinct $\approx$-classes that agree on suitable $\mathsf{fun}'$ propositions, or a pure reversible operator that is neither uniformly truth-preserving nor uniformly truth-flipping. Bacon's substitution-structure models do not already supply the desired example. The exact footnote-59 theorem now rules out the entire current tree-semantic framework whenever the PP-free M1 background and QSS are retained; M2 independently shows that identifying purity with invariance is incompatible with QSS. The operator $Z$ shows that the operators denoted by Bacon's closed logical terms already take the $\neg$L2 branch. The enlargement theorem shows that every composition-closed enlargement of those operators that admits a $\mathsf{fun}'$ proposition does so as well. The missing step is not preservation of this escape route but construction of a genuinely different Henkin interpretation that also validates PP and the remaining background theory.

**Negative route.** The most direct negative route is the one already isolated in Goodman's notes: derive enough of L2 and one of Inv, WI, TU, or RS from the remaining principles, and apply T6. The formalization verifies the relevant T6 implications and removes the need to assume a $\mathsf{fun}'$ witness separately. The $Z$ counterexample and its enlargement theorem show that neither closed-logical purity nor merely adding further pure operators on Bacon's tree frame secures L2 under the stated closure and $\mathsf{fun}'$ conditions. Any negative proof deriving L2 from PP must therefore use features not present in these tree interpretations; alternatively, it would show that no such PP enlargement exists.

There is now a second, more direct negative route. The exact footnote-59 theorem rules out PP in Bacon's tree semantics whenever the PP-free M1 background and QSS hold. If that type-correct diagonal argument can be lifted from the tree construction to arbitrary CEV+ Henkin models, it yields a model-independent negative answer without first deriving L2.

**Relative plausibility.** The formal work does not establish that either answer is more likely. It shows exactly what each answer must overcome. A positive model must depart from Bacon's exact tree semantics and must take one of the escape routes forced by the T6 disjunction. A negative proof may either generalize the exact M1 obstruction or promote one of Goodman's model-sensitive principles, especially L2 or TU, to a consequence of the background theory itself.

# 9 Remaining verification work on the notes

The following items should not be reported as verified:

1. T7b, until a precise kind-level diagonal formula is supplied.

2. The multiple-fundamental "wide Fun" discussion in M4, which does not state a precise selection condition on Fun or a precise Separated Structure formula.

# 10 Recommended next steps

The following tasks arise directly from the unresolved or qualified claims in Goodman's notes.

1. **Generalize the exact M1 obstruction.** Abstract the type-correct footnote-59 proof from Bacon's tree semantics to an arbitrary CEV+ Henkin model. The object-language purity proof and the exact diagonal calculation are now available; the remaining task is to determine which general semantic assumptions they require. Success would answer Goodman's question negatively.

2. **Obtain determinate formulations.** Obtain an exact T7b statement, an exact formulation of the wide-Fun discussion before encoding them.

3. **Isolate any genuinely PP-specific M5 premise.** The unrestricted existential generalization of the M5 collision method is false: reversible operators are injective, and the rebuilt exotic operator is a non-truth-uniform bijection. Further work should pursue this route only if PP supplies an additional constraint on the inverse witness that is strong enough to force the displayed kind of collision.

4. **Test Goodman's negative route.** Determine whether L2 follows from PP in the full theory even though it fails throughout the relevant composition-closed enlargements on Bacon's tree frame. A positive derivation, together with TU, WI, Inv, or the strong-L2+RS route, would yield the corresponding T6 contradiction and would also explain why no such tree enlargement can validate PP. The bounded Vampire experiments show that further work should concentrate on deriving one of the three sufficient conditions isolated above, rather than repeating the monolithic refutation search.

5. **Keep model claims and proof-theoretic claims separate.** In particular, do not promote the conditional M4/M6 stock theorems to claims about Bacon's appendix model until their stock premises are instantiated.

# 11 Reproducibility

The repository's three-layer organization and current build commands are documented in `README.md` and `docs/REPOSITORY_STRUCTURE.md`. We do not reproduce that implementation documentation here.

The controlling verification ledger is

`reports/GOODMAN_COMPLETE_VERIFICATION_MATRIX_2026-07-27.md`.

The dedicated audit theory is

`reports/audit_goodman_complete/Audit_Goodman_Complete.thy`.

The verification matrix gives, for each claim, the exact Isabelle theorem and the qualification under which it has been proved. The project status and handoff files identify the source theory corresponding to each result.

The appropriate overall verdict is therefore: *the formalization verifies, corrects, or precisely qualifies every determinate claim in Goodman's notes. It completes the rebuilding step in M5, proves that the unrestricted existential generalization of the M5 collision method is false, refutes L2 in Bacon's appendix model and every relevant composition-closed enlargement, proves the exact T6 semantic consequence, and completes Bacon's footnote-59 denotation argument in the exact tree semantics. The last result rules out the current tree-model program, but it does not yet establish model-independent inconsistency. The scopes of QLN and Modalized Functionality are now stated exactly, and the remaining underspecified and model-dependent claims are explicitly identified. Goodman's consistency question itself remains open.*